Tata Kelola dan Manajemen Teknologi Informasi

Manajemen TI Domain APO

Team Teaching ISM

S1 Sistem Informasi – Fakultas Rekayasa Industri

Outline

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APO01 - Managed I&T **Management Framework**

Purpose: Implement a **consistent management approach** for enterprise governance requirements **to be met**.

Practice	Short Description
APO01.01 **Design** the management **system** for enterprise I&T	Design a management system tailored to the needs of the enterprise.
APO01.02 **Communicate** management **objectives, direction and decisions** made	Communicate awareness and promote understanding of alignment and I&T objectives to stakeholders throughout the enterprise.
APO01.03 **Implement** management **processes**	Define target process capability levels and implementation priority based on the management system design.
APO01.04 Define and implement the **organizational structures**	Put in place the required internal and extended organizational structures (e.g., committees) per the management system design, enabling effective and efficient decision making.
APO01.05 Establish **roles and responsibilities**	Define and communicate roles and responsibilities for enterprise I&T, including authority levels, responsibilities and accountability.
APO01.06 Optimize the **placement** of the **IT function**	Position the IT capabilities in the overall organizational structure to reflect the strategic importance and operational dependency of IT within the enterprise.
APO01.07 Define information **(data) and system ownership**	Define and maintain responsibilities for ownership of information (data) and information systems.
APO01.08 Define **target skills** and **competencies**	Define the required skills and competencies to achieve relevant management objectives.
APO01.09 Define and communicate **policies and procedures**	Put in place procedures to maintain compliance with and performance measurement of policies and other components of the control framework.

APO01 - Managed I&T **Management Framework**

Purpose: Implement a **consistent management approach** for enterprise governance requirements **to be met**.

Practice	Short Description
APO01.10 Define and implement **infrastructure, services and applications**	Define and implement infrastructure, services and applications to support the governance and management system.
APO01.11 Manage **continual improvement** of the I&T management system	Continually improve processes and other management system components.

APO01 - Managed I&T **Management Framework**

Purpose: Implement a **consistent management approach** for enterprise governance requirements **to be met**.

B. Component: Organizational Structures																					
Key Management Practice	**Executive Committee**	**Chief Risk Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**I&T Governance Board**	**Architecture Board**	**Enterprise Risk Committee**	**Chief Information Security Officer**	**Business Process Owners**	**Data Management Function**	**Head Human Resources**	**Relationship Manager**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
APO01.01 Design the management system for enterprise I&T.	A		R	R	R	R															
APO01.02 Communicate management objectives, direction and decisions made.	A	R	R	R	R	R			R				R								
APO01.03 Implement management processes (to support the achievement of governance and management objectives).	A	R	R	R	R	R			R												
APO01.04 Define and implement the organizational structures.	A		R	R	R	R						R									
APO01.05 Establish roles and responsibilities.	A		R	R	R	R															
APO01.06 Optimize the placement of the IT function.	A		R	R	R	R		R													
APO01.07 Define information (data) and system ownership.	A		R	R	R	R		R		R	R			R							
APO01.08 Define target skills and competencies.	A		R	R	R	R								R	R	R	R				
APO01.09 Define and communicate policies and procedures.	A		R	R	R	R	R	R		R	R	R		R	R	R	R	R	R	R	R
APO01.10 Define and implement infrastructure, services and applications to support the governance and management system.	A		R	R	R	R					R			R	R	R	R	R	R	R	R
APO01.11 Manage continual improvement of the I&T management system.	A		R	R	R	R				R	R			R	R	R	R	R	R	R	R

APO02 - Managed **Strategy**

Purpose: **Support** the digital transformation strategy of the organization and **deliver** the desired value **through** a road map of incremental changes.

Practice	Short Description
APO02.01 Understand enterprise **context** and **direction**	Understand the enterprise context (industry drivers, relevant regulations, basis for competition), its current way of working and its ambition level in terms of digitization.
APO02.02 Assess **current** capabilities, performance and digital maturity of the enterprise	Assess the performance of current I&T services and develop an understanding of current business and I&T capabilities (both internal and external).
APO02.03 Define **target** digital capabilities	Based on the understanding of enterprise context and direction, define the target I&T products and services and required capabilities.
APO02.04 Conduct a **gap analysis**	Identify gaps between current and target environments and describe the high-level changes in the enterprise architecture.
APO02.05 Define the **strategic plan** and **road map**	Develop a holistic digital strategy, in cooperation with relevant stakeholders, and detail a road map that defines the incremental steps required to achieve the goals and objectives.
APO02.06 **Communicate** the I&T strategy and direction	Create awareness and understanding of the business and I&T objectives and direction, as captured in the I&T strategy, through communication to appropriate stakeholders and users throughout the enterprise.

APO02 - Managed **Strategy**

Purpose: **Support** the digital transformation strategy of the organization and **deliver** the desired value **through** a road map of incremental changes.

B. Component: Organizational Structures																	
Key Management Practice	**Chief Executive Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**I&T Governance Board**	**Business Process Owners**	**Project Management Office**	**Data Management Function**	**Relationship Manager**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
APO02.01 Understand enterprise context and direction.		A	R	R				R	R	R	R	R	R	R	R	R	R
APO02.02 Assess current capabilities, performance and digital maturity of the enterprise.		A	R	R				R		R	R	R	R	R	R	R	R
APO02.03 Define target digital capabilities.		R	R	A		R		R	R	R	R	R	R	R	R	R	R
APO02.04 Conduct a gap analysis.		R	R	R	A	R		R		R	R	R	R	R	R	R	R
APO02.05 Define the strategic plan and road map.		R	R	R	A	R	R	R		R	R	R	R	R	R	R	R
APO02.06 Communicate the I&T strategy and direction.	R	R	R	R	A												

APO03 - Managed **Enterprise Architecture**

Purpose: **Represent** the different building blocks that make up the enterprise and its interrelationships as well as the principles guiding their design and evolution over time.

Practice	Short Description
APO03.01 Develop the enterprise **architecture vision**	The architecture vision provides a first-cut, high-level description of the baseline and target architectures, covering the business, information, data, application and technology domains.
APO03.02 Define **reference architecture**	The reference architecture describes the current and target architectures for the business, information, data, application and technology domains.
APO03.03 Select **opportunities** and **solutions**	Rationalize the gaps between baseline and target architectures, accounting for both business and technical perspectives, and logically group them into project work packages.
APO03.04 **Define** architecture **implementation**	Create a viable implementation and migration plan in alignment with the program and project portfolios.
APO03.05 **Provide** enterprise architecture **services**	Provide enterprise architecture services within the enterprise.

APO03 - Managed **Enterprise Architecture**

Purpose: **Represent** the different building blocks that make up the enterprise and its interrelationships as well as the principles guiding their design and evolution over time.

B. Component: Organizational Structures								
Key Management Practice	**Chief Operating Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**I&T Governance Board**	**Architecture Board**	**Data Management Function**	**Head Architect**
APO03.01 Develop the enterprise architecture vision.		R	R	R	R	A	R	R
APO03.02 Define reference architecture.		R	R	R	R	A	R	R
APO03.03 Select opportunities and solutions.		R	R	R	R	A	R	R
APO03.04 Define architecture implementation.	R	R	R	R	R	A	R	R
APO03.05 Provide enterprise architecture services.	R	R	R	R	R	A		R

APO04 - Managed **Innovation**

Purpose: **Achieve** competitive advantage, business innovation, improved customer experience, and improved operational effectiveness and efficiency **by exploiting** I&T developments and emerging technologies.

Practice	Short Description
APO04.01 Create an **environment conducive** to innovation	Create an environment that is conducive to innovation, considering methods such as culture, reward, collaboration, technology forums, and mechanisms to promote and capture employee ideas.
APO04.02 Maintain an understanding of the **enterprise environment**	Maintain an adequate understanding of enterprise strategy, competitive environment and other constraints, so that opportunities enabled by new technologies can be identified.
APO04.03 Monitor and scan the **technology environment**	Set up a technology watch process to perform systematic monitoring and scanning of the enterprise's external environment to identify emerging technologies that have the potential to create value.
APO04.04 **Assess** the potential of **emerging technologies** and **innovative ideas**	Analyze identified emerging technologies and/or other I&T innovative suggestions to understand their business potential.
APO04.05 **Recommend** appropriate further **initiatives**	Evaluate and monitor the results of proof-of-concept initiatives and, if favorable, generate recommendations for further initiatives. Gain stakeholder support.
APO04.06 **Monitor** the implementation and use of innovation	Monitor the implementation and use of emerging technologies and innovations to ensure that the promised benefits are realized and to identify lessons learned.

APO04 - Managed **Innovation**

Purpose: **Achieve** competitive advantage, business innovation, improved customer experience, and improved operational effectiveness and efficiency **by exploiting** I&T developments and emerging technologies.

B. Component: Organizational Structures													
Key Management Practice	Executive Committee	Chief Information Officer	Chief Technology Officer	Chief Digital Officer	Business Process Owners	Data Management Function	Head Human Resources	Relationship Manager	Head Architect	Head Development	Head IT Operations	Service Manager	Information Security Manager
APO04.01 Create an environment conducive to innovation.	A	R	R	R	R	R	R		R	R	R	R	R
APO04.02 Maintain an understanding of the enterprise environment.	A	R	R	R	R	R		R	R	R	R	R	R
APO04.03 Monitor and scan the technology environment.	A	R	R	R	R	R			R	R	R	R	R
APO04.04 Assess the potential of emerging technologies and innovative ideas.	A	R	R	R	R	R			R	R	R	R	R
APO04.05 Recommend appropriate further initiatives.	A	R	R	R	R	R			R	R	R	R	R
APO04.06 Monitor the implementation and use of innovation.	A	R	R	R	R	R			R	R	R	R	R

APO05 - Managed **Portfolio**

Purpose: **Optimize** the **performance** of the overall portfolio of programs in response to individual program, product and service performance and changing enterprise priorities and demand.

Practice	Short Description
APO05.01 Determine the availability and **sources of funds**	Determine potential sources of funds, different funding options and the implications of the funding source on the investment return expectations.
APO05.02 Evaluate and **select programs** to fund	Based on requirements for the overall investment portfolio mix and the I&T strategic plan and road map, evaluate and prioritize program business cases and decide on investment proposals.
APO05.03 **Monitor, optimize and report** on investment portfolio **performance**	On a regular basis, monitor and optimize the performance of the investment portfolio and individual programs throughout the entire investment life cycle.
APO05.04 Maintain **portfolios**	Maintain portfolios of investment programs and projects, I&T products and services, and I&T assets.
APO05.05 Manage benefits **achievement**	Monitor the benefits of providing and maintaining appropriate I&T products, services and capabilities, based on the agreed and current business case.

APO05 - Managed **Portfolio**

Purpose: **Optimize** the **performance** of the overall portfolio of programs in response to individual program, product and service performance and changing enterprise priorities and demand.

B. Component: Organizational Structures									
Key Management Practice	**Chief Financial Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**I&T Governance Board**	**Business Process Owners**	**Portfolio Manager**	**Program Manager**	**Project Management Office**
APO05.01 Determine the availability and sources of funds.	R	R			A		R		
APO05.02 Evaluate and select programs to fund.	R	R	R	R	A		R	R	
APO05.03 Monitor, optimize and report on investment portfolio performance.		R	R	R	A		R	R	
APO05.04 Maintain portfolios.		R	R	R	A		R	R	R
APO05.05 Manage benefits achievement.	R	R	R	R	A	R	R	R	

APO06 - Managed **Budget and Costs**

Purpose: Foster a partnership between IT and enterprise stakeholders to **enable the effective and efficient use** of I&T-related resources and **provide transparency and accountability** of the cost and business value of solutions and services.

Practice	Short Description
APO06.01 Manage **finance and accounting**	Establish and maintain a method to manage and account for all I&T-related costs, investments and depreciation as an integral part of enterprise financial systems and accounts.
APO06.02 Prioritize **resource allocation**	Implement a decision-making process to prioritize the allocation of resources and establish rules for discretionary investments by individual business units.
APO06.03 Create and maintain **budgets**	Prepare a budget reflecting investment priorities based on the portfolio of I&T-enabled programs and I&T services.
APO06.04 Model and **allocate costs**	Establish and use an I&T costing model based, for example, on the service definition.
APO06.05 **Manage costs**	Implement a cost management process that compares actual costs against budget.

APO06 - Managed **Budget and Costs**

Purpose: Foster a partnership between IT and enterprise stakeholders to **enable the effective and efficient use** of I&T-related resources and **provide transparency and accountability** of the cost and business value of solutions and services.

B. Component: Organizational Structures						
Key Management Practice	**Chief Financial Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**Portfolio Manager**	**Head IT Administration**
APO06.01 Manage finance and accounting.	A				R	R
APO06.02 Prioritize resource allocation.	R	A	R	R	R	R
APO06.03 Create and maintain budgets.	R	A	R	R		R
APO06.04 Model and allocate costs.	R	A				R
APO06.05 Manage costs.	R	A	R	R		R

APO07 - Managed **Human Resources**

Purpose: **Optimize** human resources capabilities to **meet** enterprise objectives.

Practice	Short Description
APO07.01 Acquire and maintain adequate and appropriate **staffing**	Establish and maintain a method to manage and account for all I&T-related costs, investments and depreciation as an integral part of the enterprise financial systems and accounts.
APO07.02 Identify **key IT personnel**	Identify key IT personnel. Use knowledge capture (documentation), knowledge sharing, succession planning and staff backup to minimize reliance on a single individual performing a critical job function.
APO07.03 **Maintain the skills and competencies** of personnel	Define and manage the skills and competencies required of personnel. Regularly verify that personnel have the competencies to fulfill their roles on the basis of their education, training and/or experience.
APO07.04 **Assess** and recognize/**reward** employee **job performance**	Conduct timely, regular performance evaluations against individual objectives derived from enterprise goals, established standards, specific job responsibilities, and the skills and competency framework.
APO07.05 **Plan and track** the usage of **IT and business human resources**	Understand and track the current and future demand for business and IT human resources with responsibilities for enterprise I&T.
APO07.06 Manage **contract staff**	Ensure that consultants and contract personnel who support the enterprise with I&T skills know and comply with the organization's policies and meet agreed contractual requirements.

APO07 - Managed **Human Resources**

Purpose: **Optimize** human resources capabilities to **meet** enterprise objectives.

B. Component: Organizational Structures																	
Key Management Practice	**Chief Financial Officer**	**Chief Operating Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**Project Management Office**	**Head Human Resources**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**	**Legal Counsel**
APO07.01 Acquire and maintain adequate and appropriate staffing.			A	R	R	R	R	R	R	R	R	R	R	R		
APO07.02 Identify key IT personnel.			A	R	R	R	R	R	R	R	R	R	R	R	R	R
APO07.03 Maintain the skills and competencies of personnel.			A	R	R	R	R	R	R	R	R	R	R	R		
APO07.04 Assess and recognize/reward employee job performance.			A			R	R	R	R	R	R	R	R	R		
APO07.05 Plan and track the usage of IT and business human resources.	R	A	R	R	R	R	R	R	R	R	R	R	R	R		
APO07.06 Manage contract staff.			A	R	R	R	R	R	R	R	R	R	R	R		R

APO08 - Managed **Relationships**

Purpose: Enable the right knowledge, skills and behaviors to **create** improved outcomes, increased confidence, mutual trust and effective use of resources that **stimulate** a productive relationship with business stakeholders.

Practice	Short Description
APO08.01 Understand **business expectations**	Understand current business issues, objectives and expectations for I&T.
APO08.02 **Align** I&T strategy with business expectations and **identify opportunities** for IT to enhance the business	Align I&T strategies with current business objectives and expectations to enable IT to be a value-add partner for the business and a governance component for enhanced enterprise performance.
APO08.03 **Manage** the business **relationship**	Manage the relationship between the IT service organization and its business partners. Ensure that relationship roles and responsibilities are defined and assigned, and communication is facilitated.
APO08.04 **Coordinate and communicate**	Work with all relevant stakeholders and coordinate the end-to-end delivery of I&T services and solutions provided to the business.
APO08.05 **Provide input** to the **continual improvement** of services.	Continually improve and evolve I&T-enabled services and service delivery to the enterprise to align with changing enterprise objectives and technology.

APO08 - Managed **Relationships**

Purpose: Enable the right knowledge, skills and behaviors to **create** improved outcomes, increased confidence, mutual trust and effective use of resources that **stimulate** a productive relationship with business stakeholders.

B. Component: Organizational Structures																
Key Management Practice	**Chief Executive Officer**	**Chief Financial Officer**	**Chief Operating Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**I&T Governance Board**	**Business Process Owners**	**Relationship Manager**	**Head Architect**	**Head Development**	**Head IT Operations**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
APO08.01 Understand business expectations.				A	R	R		R	R		R	R	R	R	R	R
APO08.02 Align I&T strategy with business expectations and identify opportunities for IT to enhance the business.				A	R	R	R	R	R	R	R	R	R			
APO08.03 Manage the business relationship.	R	R	R	A	R	R		R	R		R	R	R			
APO08.04 Coordinate and communicate.	R	R	R	A	R	R		R	R		R	R	R			
APO08.05 Provide input to the continual improvement of services.				A	R	R		R	R		R	R	R			

APO09 - Managed **Service Agreements**

Purpose: **Ensure** that I&T products, services and service levels **meet** current and future enterprise needs.

Practice	Short Description
APO09.01 **Identify** I&T services	Analyze business requirements and the degree to which I&T-enabled services and service levels support business processes.
APO09.02 **Catalog** I&T-enabled services	Define and maintain one or more service catalogues for relevant target groups.
APO09.03 Define and prepare **service agreements**	Define and prepare service agreements based on options in the service catalogues.
APO09.04 Monitor and report **service levels**	Monitor service levels, report on achievements and identify trends. Provide the appropriate management information to aid performance management.
APO09.05 **Review** service **agreements and contracts**	Conduct periodic reviews of the service agreements and revise when needed.

APO09 - Managed **Service Agreements**

Purpose: **Ensure** that I&T products, services and service levels **meet** current and future enterprise needs.

B. Component: Organizational Structures											
Key Management Practice	**Chief Operating Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Enterprise Risk Committee**	**Business Process Owners**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Privacy Officer**	**Legal Counsel**
APO09.01 Identify I&T services.	R	R	A		R			R			
APO09.02 Catalog I&T-enabled services.		R	A	R				R			
APO09.03 Define and prepare service agreements.		R	A			R	R	R	R	R	R
APO09.04 Monitor and report service levels.		R	A		R			R			R
APO09.05 Review service agreements and contracts.	R	A	R			R	R	R			

APO10 - Managed **Vendors**

Purpose: Optimize available I&T capabilities to support the I&T strategy and road map, **minimize the risk** associated with nonperforming or noncompliant vendors, and **ensure** competitive pricing.

Practice	Short Description
APO10.01 **Identify and evaluate** vendor **relationships and contracts**	Continuously search for and identify vendors and categorize them into type, significance and criticality.
APO10.02 **Select** vendors	Select suppliers according to a fair and formal practice to ensure a viable best fit based on specified requirements.
APO10.03 **Manage** vendor **relationships and contracts**	Formalize and manage the supplier relationship for each supplier. Manage, maintain and monitor contracts and service delivery.
APO10.04 Manage vendor **risk**	Identify and manage risk relating to vendors' ability to continually provide secure, efficient and effective service delivery.
APO10.05 Monitor vendor **performance and compliance**	Periodically review overall vendor performance, compliance to contract requirements and value for money. Address identified issues.

APO10 - Managed **Vendors**

Purpose: Optimize available I&T capabilities to support the I&T strategy and road map, **minimize the risk** associated with nonperforming or noncompliant vendors, and **ensure** competitive pricing.

B. Component: Organizational Structures													
Key Management Practice	**Chief Risk Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**I&T Governance Board**	**Enterprise Risk Committee**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Privacy Officer**	**Legal Counsel**
APO10.01 Identify and evaluate vendor relationships and contracts.		R	R	R	A				R				R
APO10.02 Select vendors.		R	R	R	A		R	R	R	R	R	R	
APO10.03 Manage vendor relationships and contracts.		R	R	R	A		R	R	R	R			R
APO10.04 Manage vendor risk.	R	R	R	R	A	R	R	R	R	R	R	R	
APO10.05 Monitor vendor performance and compliance.	R	R	R	R	A	R	R	R	R	R			R

APO11 - Managed **Quality**

Purpose: Ensure **consistent delivery** of technology solutions and services to **meet** the quality requirements of the enterprise and **satisfy** stakeholder needs.

Practice	Short Description
APO11.01 **Establish** a quality management **system** (QMS)	Establish and maintain a quality management system (QMS) that provides a standard, formal and continuous approach to quality management of information.
APO11.02 **Focus** quality management **on customers**	Focus quality management on customers by determining their requirements and ensuring integration in quality management practices.
APO11.03 Manage quality **standards, practices and procedures** and integrate quality management into key processes and solutions	Identify and maintain standards, procedures and practices for key processes to guide the enterprise in meeting the intent of the agreed quality management standards (QMS).
APO11.04 Perform quality **monitoring, control and reviews**	Monitor the quality of processes and services on an ongoing basis, in line with quality management standards.
APO11.05 Maintain **continuous improvement**	Maintain and regularly communicate an overall quality plan that promotes continuous improvement.

APO11 - Managed **Quality**

Purpose: Ensure **consistent delivery** of technology solutions and services to **meet** the quality requirements of the enterprise and **satisfy** stakeholder needs.

B. Component: Organizational Structures																				
Key Management Practice	**Chief Operating Officer**	**Chief Risk Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**I&T Governance Board**	**Business Process Owners**	**Portfolio Manager**	**Program Manager**	**Project Manager**	**Project Management Office**	**Data Management Function**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**
APO11.01 Establish a quality management system (QMS).	A		R		R											R	R		
APO11.02 Focus quality management on customers.			A		R		R										R		
APO11.03 Manage quality standards, practices and procedures and integrate quality management into key processes and solutions.			A	R	R		R	R	R	R	R	R	R	R	R	R	R	R	R
APO11.04 Perform quality monitoring, control and reviews.		R	A		R	R	R										R		
APO11.05 Maintain continuous improvement.			A				R	R	R	R	R		R	R	R	R	R	R	R

APO12 - Managed **Risk**

Purpose: **Integrate** the management of I&T-related enterprise risk with overall enterprise risk management (ERM) and balance the costs and benefits of managing I&T-related enterprise risk.

Practice	Short Description
APO12.01 **Collect** data	Identify and collect relevant data to enable effective I&T-related risk identification, analysis and reporting.
APO12.02 **Analyze** risk	Develop a substantiated view on actual I&T risk, in support of risk decisions.
APO12.03 **Maintain** a risk profile	Maintain an inventory of known risk and risk attributes, including expected frequency, potential impact and responses.
APO12.04 **Articulate** risk	Communicate information on the current state of I&T-related exposures and opportunities in a timely manner to all required stakeholders for appropriate response.
APO12.05 Define a risk management **action portfolio**	Manage opportunities to reduce risk to an acceptable level as a portfolio.
APO12.06 **Respond** to risk	Respond in a timely manner to materialized risk events with effective measures to limit the magnitude of loss.

APO12 - Managed **Risk**

Purpose: **Integrate** the management of I&T-related enterprise risk with overall enterprise risk management (ERM) and balance the costs and benefits of managing I&T-related enterprise risk.

B. Component: Organizational Structures																	
Key Management Practice	**Chief Risk Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**Enterprise Risk Committee**	**Chief Information Security Officer**	**Business Process Owners**	**Project Management Office**	**Data Management Function**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
APO12.01 Collect data.	A	R	R	R		R	R	R	R	R	R	R	R	R	R	R	R
APO12.02 Analyze risk.	A	R			R		R										
APO12.03 Maintain a risk profile.	A	R			R		R										
APO12.04 Articulate risk.	A	R			R		R										
APO12.05 Define a risk management action portfolio.	A	R			R		R										
APO12.06 Respond to risk.	R	A	R	R		R	R	R		R	R	R	R	R	R	R	R

APO13 - Managed **Security**

Purpose: **Keep the impact and occurrence** of information security incidents **within** the enterprise's risk appetite levels.

Practice	Short Description
APO13.01 **Establish** and maintain an information security management **system** (ISMS)	Establish and maintain an information security management system (ISMS) that provides a standard, formal and continuous approach to information security management, enabling secure technology and business processes that are aligned with business requirements.
APO13.02 Define and manage an information security and privacy **risk treatment plan**	Maintain an information security plan that describes how information security risk is to be managed and aligned with enterprise strategy and enterprise architecture.
APO13.03 **Monitor and review** the information security management **system** (ISMS)	Maintain and regularly communicate the need for, and benefits of, continuous improvement in information security. Collect and analyze data about the information security management system (ISMS), and improve its effectiveness. Correct nonconformities to prevent recurrence.

APO13 - Managed **Security**

Purpose: **Keep the impact and occurrence** of information security incidents **within** the enterprise's risk appetite levels.

B. Component: Organizational Structures															
Key Management Practice	**Chief Information Officer**	**Chief Technology Officer**	**Enterprise Risk Committee**	**Chief Information Security Officer**	**Business Process Owners**	**Project Management Office**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
APO13.01 Establish and maintain an information security management system (ISMS).	R		R	A						R		R		
APO13.02 Define and manage an information security and privacy risk treatment plan.	R		R	A						R		R		R
APO13.03 Monitor and review the information security management system (ISMS).	R	R		A	R	R	R	R	R	R	R	R	R	R

APO14 - Managed **Data**

Purpose: Ensure **effective utilization** of the critical data assets to achieve enterprise goals and objectives.

Practice	Short Description
APO14.01 Define and communicate the organization's data management **strategy** and **roles and responsibilities**	Define how to manage and improve the organization's data assets, in line with enterprise strategy and objectives.
APO14.02 Define and maintain a consistent **business glossary**	Create, approve, update and promote consistent business terms and definitions to foster shared data usage across the organization.
APO14.03 Establish the processes and infrastructure for **metadata management**	Establish the processes and infrastructure for specifying and extending metadata about the organization's data assets, fostering and supporting data sharing, ensuring compliant use of data, improving responsiveness to business changes and reducing data-related risk.
APO14.04 Define a **data quality strategy**	Define an integrated, organizationwide strategy to achieve and maintain the level of data quality required to support the business goals and objectives.
APO14.05 Establish **data profiling methodologies, processes and tools**	Implement standardized data profiling methodologies, processes, practices, tools and templates that can be applied across multiple data repositories and data stores.
APO14.06 Ensure a **data quality assessment approach**	Provide a systematic approach to measure and evaluate data quality according to processes and techniques, and against data quality rules.
APO14.07 Define the **data cleansing approach**	Define the mechanisms, rules, processes, and methods to validate and correct data according to predefined business rules.
APO14.08 Manage the **life cycle** of data assets	Ensure that the organization understands, maps, inventories and controls its data flows through business processes over the data life cycle, from creation or acquisition to retirement.

APO14 - Managed **Data**

Purpose: Ensure **effective utilization** of the critical data assets to achieve enterprise goals and objectives.

Practice	Short Description
APO14.09 Support **data archiving and retention**	Ensure that data maintenance satisfies organizational and regulatory requirements for availability of historical data.
APO14.10 Manage **data backup and restore arrangements**	Manage availability of critical data to ensure operational continuity.

APO14 - Managed **Data**

Purpose: Ensure **effective utilization** of the critical data assets to achieve enterprise goals and objectives.

B. Component: Organizational Structures							
Key Management Practice	**Chief Risk Officer**	**Chief Information Officer**	**Chief Digital Officer**	**Enterprise Risk Committee**	**Chief Information Security Officer**	**Data Management Function**	**Legal Counsel**
APO14.01 Define and communicate the organization's data management strategy and roles and responsibilities.	R	A	R		R	R	
APO14.02 Define and maintain a consistent business glossary.	R	A	R		R	R	
APO14.03 Establish the processes and infrastructure for metadata management.	R	A	R		R	R	
APO14.04 Define a data quality strategy.	R	A	R		R	R	
APO14.05 Establish data profiling methodologies, processes and tools.	R	A	R		R	R	
APO14.06 Ensure a data quality assessment approach.	R	A	R		R	R	
APO14.07 Define the data cleansing approach.	R	A	R		R	R	
APO14.08 Manage the life cycle of data assets.	R	A	R	R	R	R	R
APO14.09 Support data archiving and retention.	R	A	R	R	R	R	R
APO14.10 Manage data backup and restore arrangements.	R	A	R		R	R	R

Information Input Example of APO Domain

Information Input Example of APO Domain

Objective	Practice	Input	Description
APO01 - Managed I&T **Management Framework**	APO01.02 Communicate management objectives, direction and decisions made	EDM01.02	Enterprise **governance communication**
APO02 - Managed **Strategy**	APO02.01 Understand enterprise context and direction	Outside COBIT	**SWOT analysis**
APO03 - Managed **Enterprise Architecture**	APO03.01 Develop the enterprise APO03.01 architecture vision	APO02.05	Strategic **road map**
APO04 - Managed **Innovation**	APO04.03 Monitor and scan the technology environment	Outside COBIT	**Emerging technologies**
APO05 - Managed **Portfolio**	APO05.02 Evaluate and select programs to fund	APO06.03	**IT budget**
APO06 - Managed **Budget and Costs**	APO06.02 Prioritize resource allocation	APO05.02	Program **business case**
APO07 - Managed **Human Resources**	APO07.04 Assess and recognize/reward employee job performance	APO04.01	**Recognition and reward program**
APO08 - Managed **Relationships**	APO08.03 Manage the business relationship	DSS02.06	User confirmation of **satisfactory fulfilment** or resolution
APO09 - Managed **Service Agreements**	APO09.03 Define and prepare service agreements	APO11.02	Customer **requirements for quality** management
APO10 - Managed **Vendors**	APO10.01 Identify and evaluate vendor relationships and contracts	Outside COBIT	Vendor **contracts**

Information Input Example of APO Domain

Objective	Practice	Input	Description
APO11 - Managed **Quality**	APO11.03 Manage quality standards, practices and procedures and integrate quality management into key processes and solutions	Outside COBIT	Industry **good practices**
APO12 - Managed **Risk**	APO12.01 Collect data	DSS02.07	**Incident status and trends** report
APO13 - Managed **Security**	APO13.01 Establish and maintain an information security management system (ISMS)	Outside COBIT	Enterprise **security approach**
APO14 - Managed **Data**	APO14.01 Define and communicate the organization's data management strategy and roles and responsibilities.	APO01.06	**Data classification** guidelines

Thank You

That’s all